

Collecting Numbers II

Time limit: 1.00 s

Memory limit: 512 MB

You are given an array that contains each number between $1 \dots n$ exactly once. Your task is to collect the numbers from 1 to n in increasing order.

On each round, you go through the array from left to right and collect as many numbers as possible.

Given m operations that swap two numbers in the array, your task is to report the number of rounds after each operation.

Input

The first line has two integers n and m : the array size and the number of operations.

The next line has n integers $x_1, x_2, \dots, x_n$: the numbers in the array.

Finally, there are m lines that describe the operations. Each line has two integers a and b : the numbers at positions a and b are swapped.

Output

Print m integers: the number of rounds after each swap.

Constraints

- $1 \leq n, m \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$

Example

Input	Output
5 3 4 2 1 5 3 2 3 1 5 2 3	2 3 4